Tyrell Freeman

Cybersecurity Graduate Student | Governance, Risk & Compliance
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PROFESSIONAL SUMMARY

CompTIA Security+ certified U.S. Air Force Veteran with hands-on experience in third-party risk assessment, vendor lifecycle management, compliance framework application, and security documentation. Proficient in NIST SP 800-53, NIST SP 800-30, and ISO 27001, with demonstrated ability to conduct risk assessments, produce vendor risk reports, perform control gap analyses, and coordinate remediation plans. Applied the NIST Cybersecurity Framework, OWASP, CIS Controls, and MITRE ATT&CK across graduate coursework in risk management, secure scripting, cryptography, cloud security, and intrusion detection. Previously held DoD Secret Security Clearance. Seeking a GRC internship to apply framework knowledge and analytical skills in a professional environment.

CORE SKILLS & COMPETENCIES

GRC Frameworks:	NIST CSF, NIST SP 800-53, NIST SP 800-30, ISO 27001, OWASP Top 10, CIS Controls, MITRE ATT&CK, CISA Secure by Design, CVSS v3
GRC Competencies:	Risk assessment, control gap analysis, vendor lifecycle management, third-party risk, remediation planning, audit documentation, risk registers
Cloud & Security Tools:	AWS CloudTrail, CloudWatch, EventBridge, Config, Lambda, IAM, VPC; Nessus, OpenVAS, Wireshark, Rapid7 InsightIDR, Cisco Splunk, Snort
Technical Skills:	Python, PowerShell, Linux CLI, Kali Linux, Burp Suite, Nmap, AES-GCM, X.509 PKI, Microsoft Excel (Pivot Tables)
Regulatory Awareness:	HIPAA, CISA advisories, FBI/HHS joint cybersecurity guidance, PKI certificate standards, DoD security protocols
Documentation:	Risk registers, vendor risk reports, control gap analyses, executive summaries, lab assessments, technical writing

GRC PROJECTS

RISK & COMPLIANCE MANAGEMENT

Third-Party Risk Assessment & Security Documentation

Graduate Coursework & Professional Application | St. John's University

Conducted structured vendor risk assessments using NIST SP 800-30 methodology, producing formal GRC deliverables including risk registers, vendor risk reports, control gap analyses, and remediation tracking plans aligned with compliance and audit standards.

- Produced risk analysis matrices categorizing threats by likelihood, impact, and residual risk level using NIST SP 800-30 risk assessment methodology.
- Mapped identified vulnerabilities to NIST SP 800-53 and ISO 27001 control families, developing written remediation recommendations for technical and executive stakeholders.
- Applied CVSS v3 scoring and key risk indicators to prioritize findings from Nessus and OpenVAS scans; organized

vendor data and risk tracking in Microsoft Excel.

- Produced GRC documentation including risk registers, vendor risk reports, control gap analyses, and scan summaries aligned with compliance and audit reporting standards.

Tools / Frameworks: NIST SP 800-30 · NIST SP 800-53 · ISO 27001 · CVSS v3 · Nessus · OpenVAS · Excel

SIEM Platform Evaluation: Rapid7 InsightIDR vs. Cisco Splunk Enterprise Security

Intrusion Detection & Analysis | St. John's University

Conducted a structured comparative evaluation of two enterprise SIEM platforms assessing detection capabilities, AI-assisted alert triage, pricing models, deployment complexity, and organizational fit — a deliverable consistent with vendor risk assessment standards.

- Assessed Rapid7's User Behavior Analytics (UBA) and Attacker Behavior Analytics (ABA) against Splunk's SPL-based correlation rules, evaluating false positive rates and analyst workflow impact.
- Evaluated AI alert triage features: Rapid7's model (trained on 4.8 trillion events, 99.93% claimed accuracy) vs. Cisco's agentic AI with MITRE ATT&CK-mapped alert summaries.
- Compared total cost of ownership across deployment sizes, identifying key pricing inflection points and scalability trade-offs relevant to GRC vendor selection decisions.
- Produced a structured evaluation report covering pros, cons, AI feature sets, and organizational fit — consistent with tool selection analysis in a GRC role.

Tools / Frameworks: Rapid7 InsightIDR · Cisco Splunk ES · MITRE ATT&CK · UBA/ABA · SIEM Evaluation

Vulnerability Assessment & Risk Analysis — Nessus Network Scan

Lab 4 Assessment | St. John's University | Fall 2025

Conducted a structured vulnerability assessment for a simulated organization (Home Inc.) using Tenable Nessus Essentials, scanning 16+ IP addresses to identify, analyze, and prioritize security vulnerabilities. Produced a formal assessment report with CVSSv3 risk scoring, business impact analysis, and a compliance-aligned remediation plan.

- Identified and analyzed four primary findings: NAT-PMP Detection (CVSSv3 7.3, High), Palo Alto GlobalProtect Agent Privilege Escalation (CVE-2024-5921, High), VLC 3.0.21 outdated software, and SMB Signing not required.
- Produced risk analysis matrices evaluating business impact, attack surface exposure, and compliance implications for each finding, consistent with GRC audit reporting standards.
- Developed prioritized remediation plans covering access control hardening, firewall ACL updates, software patching, and secure configuration enforcement with compliance framing.
- Identified SSL self-signed certificate and untrusted certificate chain vulnerabilities with certificate lifecycle management recommendations tied to regulatory best practices.

Tools / Frameworks: Tenable Nessus Essentials · CVSSv3 · CVE-2024-5921 · Risk Matrix · Remediation Planning

CLOUD SECURITY & INCIDENT RESPONSE

AWS Cloud Incident Detection & Automated Remediation

Module 13 Learning Activity | St. John's University | Spring 2026

Configured and analyzed two AWS security labs covering automated incident detection and cloud-based remediation workflows. Built a detection pipeline using CloudTrail, CloudWatch, and EventBridge, then extended it to full automated

remediation via AWS Config and Lambda. Applied the NIST Cybersecurity Framework across all five functions to analyze both architectures.

- Mapped both lab architectures to all five NIST CSF functions: CloudTrail/Config for Identify, IAM least-privilege for Protect, CloudWatch/EventBridge for Detect, SNS/Lambda for Respond, and Config timelines for Recover.
- Simulated a misconfiguration incident by adding overly permissive EC2 inbound rules and verified Lambda automatically revoked unauthorized rules within minutes of detection.
- Reviewed IAM role scoping for AwsConfigRole and AwsConfigLambdaSGRole, confirming least-privilege design — each role limited to only required permissions.
- Documented full audit trails including Config timelines, CloudWatch Logs Insights queries, and EventBridge event patterns used for detection and verification.

Tools / Frameworks: AWS CloudTrail · CloudWatch · EventBridge · Config · Lambda · IAM · NIST CSF

RansomHub Ransomware Threat Analysis — Risk & IOC Assessment

Module 12 Learning Activity | St. John's University | April 2026

Threat analysis of RansomHub, a Ransomware-as-a-Service operation that compromised over 210 organizations across healthcare, government, water utilities, and financial services. Grounded in the FBI, CISA, and HHS joint advisory (AA24-242A), analyzing the group's attack lifecycle, IOCs, and organizational impact from a risk and detection perspective.

- Documented network-based IOCs including C2 over port 443, lateral movement via RDP/SMB, Tor-based .onion communications, and data exfiltration via MEGAsync and WinSCP.
- Analyzed three actively exploited CVEs: CVE-2023-3519 (Citrix NetScaler RCE), CVE-2023-22515 (Atlassian Confluence privilege escalation), and CVE-2020-1472 ZeroLogon.
- Identified persistence mechanisms including scheduled task masquerading, HKCU registry run key modifications, and abuse of legitimate RMM tools (AnyDesk, Atera).
- Developed SIEM and EDR detection recommendations tied to VSS deletion commands, PowerShell/WMI abuse, and the double-extortion kill chain.

Tools / Frameworks: CISA Advisory AA24-242A · CVE Analysis · SIEM/EDR Detection · Threat Intelligence

CRYPTOGRAPHIC CONTROLS & COMPLIANCE

Cryptographic Control Validation: RSA & Hybrid Encryption Compliance

RSA Cryptography Module Assessment | St. John's University

Assessed RSA implementation configurations against NIST SP 800-53 SC-13 compliance requirements. Evaluated cryptographic padding schemes, attack vectors, and hybrid encryption architecture, producing a formal assessment with NIST control citations and risk-based recommendations.

- Validated use of OAEP padding for NIST SC-13 compliance and documented why PKCS#1 v1.5 is disqualified for new federal systems — providing policy-level rationale grounded in attack history.
- Analyzed Bleichenbacher and ROBOT attack vectors to evaluate compliance and liability implications of weak cryptographic padding choices at the policy level.
- Evaluated hybrid encryption architecture (RSA-OAEP for key transport, RSA-PSS for digital authentication, AES for data encryption, HMAC for integrity verification) mapped to threat model.
- Produced a formal assessment report with NIST control citations, risk rationale, and cryptographic design recommendations aligned with federal security standards.

Tools / Frameworks: NIST SP 800-53 SC-13 · RSA-OAEP · AES · HMAC · Bleichenbacher · ROBOT Attack

Cryptographic Key Distribution & Management — PKI Security Analysis

Module 12 Assessment | CYB 625: Principles in Secure Scripting & Cryptography | St. John's University

Group research report analyzing cryptographic key distribution and Public Key Infrastructure (PKI) in enterprise environments. Examined CA trust hierarchies, mapped controls to NIST SP 800-53, and demonstrated scripted approaches to automating certificate validation and CSR generation.

- Mapped PKI implementation practices to NIST SP 800-53 controls SC-12 (Key Establishment and Management) and SC-8 (Transmission Confidentiality and Integrity).
- Analyzed certificate validation tiers (DV, OV, EV) and their appropriate use cases — demonstrating risk-proportionate control selection.
- Scripted X.509 certificate parsing in Python to automate detection of fraudulent certificates and Unicode phishing indicators.
- Identified Unicode homograph attacks as a gap in CA validation and recommended enhanced monitoring controls.

Tools / Frameworks: NIST SP 800-53 SC-12/SC-8 · X.509 PKI · Python · PowerShell · CA Trust Hierarchy

Forensic Cryptography & Evidence Integrity Controls

CYB 625 Module 8 Assessment | St. John's University

Implemented and assessed cryptographic controls for forensic evidence integrity, applying AES-GCM with Associated Authenticated Data (AAD) and Fernet symmetric encryption to secure digital evidence workflows. Connected implementation decisions to NIST SP 800-38D and chain-of-custody compliance requirements.

- Applied AES-GCM with AAD to bind forensic evidence files to specific case identifiers, ensuring tampering or reassignment causes decryption failure as a chain-of-custody control.
- Implemented Fernet symmetric encryption (AES-128-CBC + HMAC-SHA256) for authenticated data protection in investigation workflows requiring rapid deployment without sacrificing integrity.
- Connected implementation decisions to NIST SP 800-38D GCM specifications and digital evidence handling requirements.
- Produced a written lab assessment mapping each cryptographic mechanism to its compliance rationale and applicable threat model.

Tools / Frameworks: AES-GCM · AAD · Fernet · NIST SP 800-38D · HMAC-SHA256 · Chain-of-Custody Controls

FRAMEWORK APPLICATION & RISK COMMUNICATION

Network Protocol Risk Analysis: IPSec, TLS & Wireless Security

IPSec Lab & Wireless Security Lab | St. John's University

Conducted two protocol-level risk assessments analyzing encrypted network traffic and wireless security vulnerabilities, producing structured reports consistent with audit and compliance writing standards. Evaluated AWS Site-to-Site VPN architecture from a risk governance perspective.

- Analyzed 120 ESP-encrypted packets in Wireshark, identifying 8 active Security Associations by SPI value, validating anti-replay sequence numbering, and documenting findings in a formal security assessment.
- Evaluated AWS Site-to-Site VPN architecture including dual-tunnel redundancy, IKEv2 with AES-256-GCM, Dead Peer Detection, and CloudWatch monitoring from a risk and policy governance perspective.
- Performed wireless security protocol assessment comparing WEP, WPA2 (TKIP), and WPA3 (SAE) vulnerabilities;

captured live Wi-Fi traffic revealing plaintext DNS exposure and TLS handshake sequences.

- Documented protocol risk trade-offs and remediation guidance in structured reports aligned with audit and compliance documentation standards.

Tools / Frameworks: Wireshark · IPSec/IKEv2 · AES-256-GCM · WPA3/SAE · AWS VPN · TLS Analysis

Secure Scripting Techniques — Framework-Aligned Presentation

Intrusion Detection & Analysis | St. John's University

Developed and delivered a 12-slide technical presentation on mitigating security risks through secure scripting practices. Applied six authoritative security frameworks to scripting vulnerabilities and defensive countermeasures, establishing business risk context before presenting technical controls.

- Applied six frameworks: NIST SP 800-53, NIST SP 800-218 (SSDF), OWASP Top 10, CIS Benchmarks, MITRE ATT&CK T1059, and CISA Secure by Design.
- Quantified business risk using IBM Cost of a Data Breach 2024 ($4.88M average cost) and Verizon DBIR to establish risk context before presenting technical controls.
- Covered six vulnerability categories (SQL injection, XSS, insecure deserialization, hardcoded credentials, path traversal, TOCTOU) with Python, PowerShell, and shell hardening countermeasures.
- Validated all 14 references against public sources, correcting outdated NIST CSRC URLs — reflecting the diligence expected in compliance documentation.

Tools / Frameworks: NIST SP 800-53 & 800-218 · OWASP · CIS Benchmarks · MITRE ATT&CK · CISA

EDUCATION

M.S. in Cyber & Information Security — Expected: 2026

St. John's University — Queens, NY

B.B.A. in Marketing Communications — Graduated: 2020

A.S. in Information Technology Management — Graduated: 2018

Berkeley College — New York, NY

CERTIFICATIONS & TRAINING

- CompTIA Security+ (SY0-701) — In Progress
- Previously held DoD Secret Security Clearance